

Uniform q -Multinomial Transseries and Certified Inversion

Closing the $q \rightarrow 1$ endpoint gap, locating Borel singularities,
and proving a sharp least-term exponential rate

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Abstract

The combinatorial-transseries companion in `ProveIt` derives an all-order $q \rightarrow 1$ expansion of central Gaussian coefficients, but its estimates require $\tau = hx \geq \tau_0 > 0$. It explicitly leaves uniform matching to $\tau = 0$ for a different error analysis. We supply that analysis and extend it to every positive multinomial ray. A partial-fraction identity for the Bose kernel gives an exact signed remainder. After reorganizing the coefficients in powers of $1/x$, the resulting estimates hold simultaneously for every $h \geq 0$, with no restriction on hx . The error is smaller than the first omitted term and is bounded by an explicit Bernoulli expression independent of h .

We derive the meromorphic Borel transform, prove its exact Laplace-summation formula, and show that the nearest singularities in the $1/x$ Borel plane are always $\pm 2\pi i a_*$, where a_* is the smallest ray parameter and m_* below is its multiplicity. A least-term choice gives the uniform bound $2re^{-2\pi a_* x}$, retaining the modular Euler-product contribution separately. At $h = 0$ the actual remainder is asymptotic to $m_*(2\pi\sqrt{a_* x})^{-1}e^{-2\pi a_* x}$; hence its exponential rate is sharp for this truncation rule. At the resonant values $h = 2\pi/m$, the remainder instead has equivalent $m_*(2\pi m)^{-1}e^{-2\pi a_* x}$; therefore no factor tending to zero with x can multiply a bound uniform in all h . A uniform lower bound on the exact forward derivative transports these estimates to explicit inverse enclosures. Complete proofs, endpoint formulas, finite evaluation bounds, numerical checks, and ten further research directions are included.

Status. The endpoint-uniform theorem package is developed and proved here. It resolves an explicitly recorded repository gap, not a named general conjecture about transseries. The elementary kernels and the general resurgence framework have substantial prior literature. Global publication-level novelty has not been established. This is an unrefereed research draft, not a Lean-verified development.

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1 The research target and the precise contribution

1.1 What the repository already supplies

The relevant source is *Combinatorial Transseries and Their Inverses*, in the series and transseries group of the Fabius-function documents [1, 2]. Its chapter on the singular $q \rightarrow 1$ transition uses $q = e^h$, $Q = e^{-h}$ and the finite-size coordinate $\tau = hx$. It contains a dilogarithmic dominant phase, Bernoulli–polylogarithm corrections, an exact modular Euler-product factor, and formal all-order inversion formulas. Its proposition labelled `t3:prop:double-A` assumes $\tau \geq \tau_0 > 0$. The paragraph following `t3:eq:Slarge` explicitly identifies the missing uniform analysis at $\tau = 0$. The modular-factor discussion also carefully avoids claiming a full beyond-all-orders description of the Bernoulli tail.

Those are the two starting points of this article. We do not claim the repository’s dilogarithmic phase or its inverse coefficient formula as new. Nor do we infer analytic theorems from a module inventory: the present proofs are independent of the status of the repository’s formalization.

The inspected snapshot is commit

04e06e032dff1966513bfba316966d52db3646b4.

The companion’s inspected source blob is

77e494fbd30587b61b36f0e0e010c85cae3c7fa7.

The source ranges used most directly are lines 4080–4280, including the labels just specified. The provenance file in this package records the full path.

1.2 Why another finite-order expansion is not enough

Three limits must be distinguished. At fixed $h > 0$, finite-size corrections are exponentially small in x . In the interior crossover, $h \rightarrow 0$ and $hx \rightarrow \tau \in (0, \infty)$; those corrections contribute to the leading $1/h$ phase. At the classical endpoint, $h \rightarrow 0$ with $hx \rightarrow 0$ and $x \rightarrow \infty$, the coefficient functions themselves become singular in the old coordinate. A remainder constant depending on a positive lower bound for hx cannot control the third regime.

The cure is not to discard the singular coefficients. It is to absorb their powers of hx into the $1/x$ scale and to estimate the exact remainder before passing to any limit. Positivity of the multinomial weight will preserve its sign. This produces a single expansion with a rigorous meaning throughout all three regimes, even when h depends on x in an arbitrary way.

1.3 Relation to established analysis

The partial fractions for $\coth$ and the eta modular transformation are classical [5, 6]. Binet’s integral and the Stirling remainder are classical as well [7, 8]. The resurgence of Euler–Maclaurin expansions was studied by Costin and Garoufalidis [3]; Garoufalidis and Kashaev give an explicit Borel-plane description and Stokes analysis of Faddeev’s quantum dilogarithm [4]. In particular, locating a lattice of Borel poles for a q -product is not, in itself, a new phenomenon.

Our specific contribution is the combination of a positive multinomial remainder, endpoint-uniform normalization, an h -independent error envelope, the precise smallest-parameter Borel geometry, sharp endpoint and resonant remainder equivalents, and a uniform inverse certificate. The targeted literature check establishes relevant precedents, not an exhaustive priority search.

The statements below are mathematical claims with proofs; the assertion that this package resolves the stated repository gap is narrower than an assertion of worldwide priority.

Editorial note (ProveIt, 2026-09-29). *Sibling packages.* This article is one of five packages that answer the same gap of the companion for the same interpolation and are to be merged into its chapter on the singular transition $q \rightarrow 1$ [9]: this article, the central Gaussian-binomial article, the certified-inversion article, the gamma-core article, and the theta-resolved article, which continues this one (see the note after Q2 in Section 11). The five write the interpolation in different normalizations ($q = e^h$ or $Q = e^{-h}$; remainders R_K , r_K , R_M or $L_h - F_{h,K}$; smallest ray parameter a_* , α or $a_{\min}$); a normalization dictionary is left to the merge. Checked agreements: the central article's F_h is (2.5) at $r = 2$, $\mathbf{a} = (1, 1)$, and its sharp minimum $e^{-2\pi x}/(\pi\sqrt{x})$ at $\tau = 0$ is (6.8) with $m_* = 2$, $a_* = 1$; the certified-inversion article's shortest-block law has the constant of (6.8). Theorem 6.5 (fixed $h = 2\pi/m$, so $hx \rightarrow \infty$) is consistent with the central article's uniform law, which is stated only for $0 \leq hx \leq T$.

Gaussian Coefficient Calculus. Theorem `q3:thm:double-scaling` of the Fabius tree's Gaussian-coefficient volume [10] expands $\log \left[\frac{n}{k} \right]_{e^{-\tau/n}}$ to all orders only for $0 < \tau_0 \leq \tau \leq \tau_1$ and $\varepsilon \leq k/n \leq 1 - \varepsilon$: the restriction of `t3:prop:double-A`. At $r = 2$, $\mathbf{a} = (\alpha, 1 - \alpha)$, $x = n$, $\alpha n = k$, that logarithm is $F_h(n) - \kappa h n^2$ with $h = \tau/n$, so its τ is hx here. Its coefficients are $C_1 = D_1 - \tau/24$ (the term $-dh/24$ of (2.13)) and $C_{2k-1} = D_k$ for $k \geq 2$, with D_k from (2.15); ProveIt checked this at 40 digits for $k \leq 4$, $\alpha \in \{\frac{1}{5}, \frac{1}{3}, \frac{1}{2}\}$ and $\tau \in \{0.3, 1.5, 4\}$, and it follows from $\frac{d}{du} \text{Li}_s(e^{-u}) = -\text{Li}_{s-1}(e^{-u})$. Since E_K depends on the ray only through $\sum_j a_j^{-2K-1}$, Theorem 2.1, with the flat term $dM(h)$ kept, gives that expansion with a signed remainder for every $h \geq 0$, uniformly for $\varepsilon \leq \alpha \leq 1 - \varepsilon$. This article does not cite that volume.

Canonical volume and Lean. The least-term bounds of Section 6 are explicit, sharp instances of the canonical volume's general theorem `p0:thm:optimal-truncation` [11]: E_K is a Gevrey-1 remainder bound with action $2\pi a_*$ per power of $1/x$. The residual-to-root step of Theorem 7.2 is, in abstract form, the Lean theorem `Fabius.exists_eq_in_residual_interval` [12]; the special-function hypotheses (Theorem 7.1 and the forward bound) are not formalized. This article cites neither.

2 The family and the endpoint-uniform expansion

Fix $r \geq 2$ and positive real numbers $\mathbf{a} = (a_1, \dots, a_r)$. Write

$$A = \sum_{j=1}^r a_j, \quad d = r - 1, \quad \kappa = \frac{A^2 - \sum_j a_j^2}{2} > 0, \quad a_* = \min_j a_j. \quad (2.1)$$

The multiplicity of the smallest parameter is $m_* = \#\{j : a_j = a_*\}$. Also put

$$H_{\mathbf{a}} = A \log A - \sum_j a_j \log a_j > 0. \quad (2.2)$$

The positivity follows on writing $a_j = A p_j$ and using $H_{\mathbf{a}} = -A \sum_j p_j \log p_j$.

For $h > 0$, let

$$\mathcal{P}(h) = \prod_{n \geq 1} (1 - e^{-hn}), \quad (2.3)$$

$$\mathcal{A}_h(s) = \sum_{m \geq 1} \frac{e^{-ms}}{m(e^{hm} - 1)}, \quad s > 0. \quad (2.4)$$

Both converge absolutely. Define the logarithmic interpolation

$$F_h(x) = \kappa h x^2 - d \log \mathcal{P}(h) + \mathcal{A}_h(Ahx) - \sum_j \mathcal{A}_h(a_j hx), \quad x > 0. \quad (2.5)$$

At $h = 0$ define

$$F_0(x) = \log \Gamma(Ax + 1) - \sum_j \log \Gamma(a_j x + 1). \quad (2.6)$$

We prove continuity at $h = 0$ below, rather than assuming it.

If all $a_j x$ are nonnegative integers, exponentiating (2.5) gives the ordinary Gaussian multinomial coefficient

$$\exp F_h(x) = \left[\begin{matrix} Ax \\ a_1 x, \dots, a_r x \end{matrix} \right]_q, \quad q = e^h. \quad (2.7)$$

Indeed, if $P_n = \prod_{j=1}^n (1 - e^{-hj})$, then $P_n = \mathcal{P}(h) \exp \mathcal{A}_h(hn)$; extracting the powers of q from the Gaussian factorials gives the quadratic factor in (2.5). For arbitrary positive parameters, (2.5) is our specified real interpolation. Nothing in the limiting argument requires a finite field whose order tends to one.

2.1 The phase, amplitude and modular term

For $\tau > 0$ set

$$S_a(\tau) = \kappa \tau^2 + \frac{d\pi^2}{6} + \text{Li}_2(e^{-A\tau}) - \sum_j \text{Li}_2(e^{-a_j \tau}), \quad (2.8)$$

$$f(u) = \frac{1 - e^{-u}}{u}, \quad f(0) = 1, \quad (2.9)$$

$$B_a(\tau) = \frac{1}{2} \left(\log f(A\tau) - \sum_j \log f(a_j \tau) \right), \quad (2.10)$$

$$M(h) = -\log \mathcal{P}(4\pi^2/h), \quad M(0) = 0. \quad (2.11)$$

All logarithms in these real formulas are real. The leading phase satisfies $S_a(0) = 0$ and

$$S'_a(0) = H_a. \quad (2.12)$$

The modular term must not be silently discarded: it is flat as $h \downarrow 0$ but is a nonzero constant in the fixed- h , large- x problem.

Our unperturbed expression is

$$\Phi_h(x) = \frac{S_a(hx)}{h} - \frac{d}{2} \log(2\pi x) + \frac{1}{2} \log \frac{A}{\prod_j a_j} + B_a(hx) - \frac{dh}{24} + dM(h), \quad h > 0, \quad (2.13)$$

$$\Phi_0(x) = H_a x - \frac{d}{2} \log(2\pi x) + \frac{1}{2} \log \frac{A}{\prod_j a_j}. \quad (2.14)$$

For $k \geq 1$, the normalized coefficients are

$$D_k(\tau) = \frac{B_{2k}}{(2k)!} \tau^{2k-1} \left(\text{Li}_{2-2k}(e^{-A\tau}) - \sum_j \text{Li}_{2-2k}(e^{-a_j \tau}) \right). \quad (2.15)$$

Here B_{2k} denotes a Bernoulli number, not the amplitude B_a . At the endpoint the definition is continued by

$$D_k(0) = \frac{B_{2k}}{2k(2k-1)} \left(A^{1-2k} - \sum_j a_j^{1-2k} \right). \quad (2.16)$$

Define

$$T_K(h, x) = \Phi_h(x) + \sum_{k=1}^K \frac{D_k(hx)}{x^{2k-1}}, \quad R_K(h, x) = F_h(x) - T_K(h, x). \quad (2.17)$$

The empty sum defines $T_0 = \Phi_h$.

Theorem 2.1 (Endpoint-uniform signed expansion). *For every positive ray $\mathbf{a}$, every integer $K \geq 0$, every $h \geq 0$ and every $x > 0$, the expansion (2.17) obeys*

$$0 < (-1)^{K+1} R_K(h, x) < \frac{|D_{K+1}(hx)|}{x^{2K+1}} \leq E_K(x), \quad (2.18)$$

where

$$E_K(x) = \frac{|B_{2K+2}|}{2K+2} x^{-(2K+1)} \sum_{j=1}^r a_j^{-(2K+1)}. \quad (2.19)$$

The coefficient functions are smooth at $\tau = 0$, bounded on $[0, \infty)$, and tend to zero as $\tau \rightarrow \infty$. The normalized core tends to (2.14) as $h \downarrow 0$ at fixed x . Consequently the expansion is uniform along every choice $h = h(x) \geq 0$ as $x \rightarrow \infty$. In addition,

$$|\partial_x R_K(h, x)| \leq \frac{2K+2}{x} E_K(x), \quad (-1)^K \partial_x R_K(h, x) > 0. \quad (2.20)$$

The theorem is a real-axis result. It makes no assertion of uniformity in a complex sector whose Borel poles pinch the integration contour. That distinct problem is discussed in Section 11.

3 The signed kernel and the proof of uniformity

3.1 An exact remainder before asymptotic expansion

The classical partial fraction identity for $\coth$ [5] can be written, for $v > 0$, as

$$\frac{1}{e^v - 1} = \frac{1}{v} - \frac{1}{2} + 2v \sum_{\ell \geq 1} \frac{1}{(2\pi\ell)^2 + v^2}. \quad (3.1)$$

Let $b_\ell = 2\pi\ell$. A finite geometric expansion in each denominator gives

$$\frac{1}{e^v - 1} = \frac{1}{v} - \frac{1}{2} + \sum_{k=1}^K \frac{B_{2k}}{(2k)!} v^{2k-1} + r_K(v), \quad (3.2)$$

$$r_K(v) = 2(-1)^K v^{2K+1} \sum_{\ell \geq 1} \frac{b_\ell^{-2K}}{b_\ell^2 + v^2}. \quad (3.3)$$

The coefficient identity follows from

$$\frac{B_{2k}}{(2k)!} = \frac{2(-1)^{k-1} \zeta(2k)}{(2\pi)^{2k}}. \quad (3.4)$$

All series in this calculation converge absolutely for a fixed positive v . Since every denominator in (3.3) is positive,

$$0 < (-1)^K r_K(v) < \frac{|B_{2K+2}|}{(2K+2)!} v^{2K+1}. \quad (3.5)$$

No limiting interchange is needed to obtain this strict inequality.

3.2 The multinomial weight

The crucial positive function is

$$W_{\mathbf{a}}(s) = \sum_j e^{-a_j s} - e^{-As} > 0, \quad s \geq 0. \quad (3.6)$$

At zero its value is $r - 1$. For $s > 0$, every $a_j < A$, so positivity is immediate. Moreover,

$$-W'_{\mathbf{a}}(s) = \sum_j a_j e^{-a_j s} - A e^{-As} = \sum_j a_j (e^{-a_j s} - e^{-As}) > 0, \quad s > 0. \quad (3.7)$$

Substitution of (3.2) into (2.4) now yields the exact remainder identity

$$R_K(h, x) = - \sum_{m \geq 1} \frac{W_{\mathbf{a}}(mhx)}{m} r_K(hm), \quad h > 0. \quad (3.8)$$

It simultaneously proves the sign and explains why taking separate absolute values of the individual factorial remainders would be an unnecessary loss.

For completeness, the non-remainder terms assemble as follows. The $1/v$ term gives the dilogarithms divided by h ; the constant $-1/2$ gives

$$J_{\mathbf{a}}(\tau) = \frac{1}{2} \left(\log(1 - e^{-A\tau}) - \sum_j \log(1 - e^{-a_j \tau}) \right). \quad (3.9)$$

The eta modular identity [6] is

$$\mathcal{P}(h) = \sqrt{\frac{2\pi}{h}} \exp\left(-\frac{\pi^2}{6h} + \frac{h}{24}\right) \mathcal{P}(4\pi^2/h). \quad (3.10)$$

Finally,

$$J_{\mathbf{a}}(\tau) = B_{\mathbf{a}}(\tau) - \frac{d}{2} \log \tau + \frac{1}{2} \log \frac{A}{\prod_j a_j}. \quad (3.11)$$

Equations (3.10) and (3.11) give precisely the core (2.13). The finite Bernoulli terms give (2.15).

3.3 A uniform moment estimate

Lemma 3.1. *For every integer $p \geq 0$ and every $s > 0$,*

$$s^{p+1} \operatorname{Li}_{-p}(e^{-s}) \leq (p+1)!. \quad (3.12)$$

Proof. On the interval $[(m-1)s, ms]$ one has $e^{-t} \geq e^{-ms}$, and hence

$$\begin{aligned} \int_{(m-1)s}^{ms} t^p e^{-t} dt &\geq \frac{s^{p+1} e^{-ms}}{p+1} (m^{p+1} - (m-1)^{p+1}) \\ &\geq \frac{s^{p+1}}{p+1} m^p e^{-ms}. \end{aligned}$$

The second inequality follows from $(m-1)^{p+1} \leq (m-1)m^p$. Summing the integrals gives $p!$, proving the claim. $\square$

Apply (3.5) in (3.8). The resulting first-omitted term is

$$\begin{aligned} |R_K(h, x)| &< \frac{|B_{2K+2}|}{(2K+2)!} h^{2K+1} \left(\sum_j \text{Li}_{-2K}(e^{-a_j hx}) - \text{Li}_{-2K}(e^{-Ahx}) \right) \\ &= \frac{|D_{K+1}(hx)|}{x^{2K+1}}. \end{aligned} \quad (3.13)$$

Discard the negative last polylogarithm and use Lemma 3.1 at $s = a_j hx$. The result is exactly (2.19), with no dependence on h . Differentiating (3.8) is justified by exponential convergence. Equation (3.7) gives the derivative's sign. The same argument with $p = 2K + 1$ gives

$$|\partial_x R_K| \leq |B_{2K+2}| x^{-(2K+2)} \sum_j a_j^{-(2K+1)}, \quad (3.14)$$

which is (2.20).

3.4 Regularity of the coefficient functions

The function $(e^s - 1)^{-1}$ is meromorphic near $s = 0$, with principal part $1/s$. Repeated differentiation shows

$$\text{Li}_{-p}(e^{-s}) = p! s^{-p-1} + \text{a function analytic at } s = 0. \quad (3.15)$$

Multiplying by τ^{p+1} in (2.15) removes the pole and gives (2.16). The coefficient functions thus extend analytically near zero. At positive infinity they are a polynomial in τ times exponentially small functions, so they tend to zero. Continuity on compact intervals then proves boundedness.

Likewise f is analytic and nonzero near zero, and the balanced relation $A = \sum_j a_j$ cancels the linear term of its logarithm. Consequently $B_a(\tau) = O(\tau^2)$. Differentiating (2.8) gives

$$S'_a(\tau) = 2\kappa\tau + A \log(1 - e^{-A\tau}) - \sum_j a_j \log(1 - e^{-a_j\tau}). \quad (3.16)$$

Writing the logarithms as $\log(a_j\tau) + \log f(a_j\tau)$ proves (2.12). Also $M(h) \rightarrow 0$, directly from its rapidly convergent product representation. This establishes the core limit.

3.5 The classical endpoint and continuity in h

Binet's integral for $z > 0$ [7] reads

$$\log \Gamma(z + 1) = \left(z + \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \int_0^\infty \frac{e^{-zt}}{t} \left(\frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} \right) dt. \quad (3.17)$$

Combining the gamma factors and using the same finite kernel expansion yields

$$R_K(0, x) = 2(-1)^{K+1} \sum_{\ell \geq 1} b_\ell^{-2K} \int_0^\infty \frac{t^{2K} W_a(xt)}{b_\ell^2 + t^2} dt. \quad (3.18)$$

Its sign is strict. Replacing its denominator by b_ℓ^2 proves the strict first-omitted-term estimate at $h = 0$. The weaker bound $E_K(x)$ follows at once; it also follows by taking the endpoint limit in the already normalized bound. The derivative argument is identical.

There is a useful direct continuity check. Set $N = 2K + 1$ and $t = hm$ in (3.8). Its absolute value is

$$2h \sum_{m \geq 1} (hm)^{N-1} W_a(xhm) \sum_{\ell \geq 1} \frac{b_\ell^{-(N-1)}}{b_\ell^2 + (hm)^2}. \quad (3.19)$$

This is a Riemann sum for the positive integral in (3.18). The integrand is continuous at zero and has exponential decay at infinity; the ℓ -sum is locally uniform and dominated by a convergent constant series. A compact-interval Riemann-sum argument followed by an exponential tail bound therefore gives convergence to (3.18). Together with the finite coefficient and core limits, this proves $F_h(x) \rightarrow F_0(x)$ for each $x > 0$. All assertions of Theorem 2.1 are now established.

4 Explicit specializations and the three limiting regimes

4.1 Central Gaussian coefficients

For $\mathbf{a} = (1, 1)$, $A = 2$, $\kappa = 1$, $d = 1$, and

$$S(\tau) = \tau^2 + \frac{\pi^2}{6} + \text{Li}_2(e^{-2\tau}) - 2 \text{Li}_2(e^{-\tau}), \quad (4.1)$$

$$S'(\tau) = 2 \log(1 + e^\tau), \quad (4.2)$$

$$B(\tau) = \frac{1}{2} \log \left(\frac{\tau}{2} \coth \frac{\tau}{2} \right), \quad (4.3)$$

$$\Phi_h(x) = \frac{S(hx)}{h} - \frac{1}{2} \log(\pi x) + B(hx) - \frac{h}{24} + M(h). \quad (4.4)$$

Thus the repository phase is preserved exactly; the new feature is its endpoint-uniform interpretation and error law. The endpoint coefficients begin

$$D_1(0) = -\frac{1}{8}, \quad D_2(0) = \frac{1}{192}, \quad D_3(0) = -\frac{1}{640}, \quad D_4(0) = \frac{17}{14336}. \quad (4.5)$$

Accordingly,

$$\log \frac{\Gamma(2x+1)}{\Gamma(x+1)^2} \sim x \log 4 - \frac{1}{2} \log(\pi x) - \frac{1}{8x} + \frac{1}{192x^3} - \frac{1}{640x^5} + \cdots. \quad (4.6)$$

This classical expansion is not a novelty claim. It is the required endpoint of the new uniform formula.

For every $h \geq 0$, the first correction gives the rigorous enclosure

$$0 < F_h(x) - T_1(h, x) < \frac{|D_2(hx)|}{x^3} \leq \frac{1}{60x^3}. \quad (4.7)$$

The last constant is intentionally simple, not sharp: at $h = 0$ the sharper next-term bound is $1/(192x^3)$.

4.2 A genuinely noncentral ray

For $\mathbf{a} = (1, 2)$, the lattice values are $\left[\frac{3x}{x, 2x} \right]_q$ and

$$\kappa = 2, \quad H_a = 3 \log 3 - 2 \log 2, \quad D_1(0) = \frac{1}{12} \left(\frac{1}{3} - 1 - \frac{1}{2} \right) = -\frac{7}{72}. \quad (4.8)$$

The smallest parameter has multiplicity one. This changes the sharp remainder constant by a factor of two compared with the central ray, while leaving the exponential rate $2\pi x$ unchanged. For $\mathbf{a} = (1, 1, 1)$ the smallest parameter has multiplicity three, so the sharp constant is three times the single-factor one. These distinctions emerge directly from Borel residues, not from the quadratic dominant phase.

4.3 Classical, crossover, and fixed- q limits

At the classical endpoint,

$$S_{\mathbf{a}}(\tau) = H_{\mathbf{a}}\tau + \frac{\kappa}{2}\tau^2 + \frac{A^3 - \sum_j a_j^3}{72}\tau^3 + O(\tau^5). \quad (4.9)$$

The term $\kappa\tau^2/(2h)$ explains the first deviation from the classical multinomial phase. A regime such as $hx \rightarrow 0$ does not imply that $hx^2 \rightarrow 0$; retaining the complete phase avoids imposing this extra, unnecessary condition.

For $hx \rightarrow \tau \in (0, \infty)$, the terms in (2.15) are bounded functions of τ multiplying $x^{-1}, x^{-3}, \dots$. This is the interior crossover previously treated in the repository, now with explicit global constants.

For fixed $h > 0$ and $x \rightarrow \infty$, the dilogarithms and all $D_k(hx)$ decay exponentially. The core recombines by (3.10) into $\kappa hx^2 - d \log \mathcal{P}(h)$ plus the appropriate finite-size terms. Thus no spurious power correction is introduced into the fixed- q problem. The uniform bound is then often conservative, but remains valid.

5 The exact Borel transform and its smallest action

5.1 Conventions and a single tail

For a formal series without constant term use

$$\mathcal{B}\left(\sum_{n \geq 0} c_n h^{n+1}\right)(\xi) = \sum_{n \geq 0} \frac{c_n}{n!} \xi^n, \quad (\mathcal{L}\beta)(h) = \int_0^\infty e^{-\xi/h} \beta(\xi) d\xi. \quad (5.1)$$

In particular there is no extra factor $1/h$ in the Laplace transform. For $s > 0$ define

$$\beta_s(\xi) = \sum_{\ell \geq 1} \frac{1}{b_\ell^2} \left(\frac{1}{\exp(s - i\xi/b_\ell) - 1} + \frac{1}{\exp(s + i\xi/b_\ell) - 1} \right). \quad (5.2)$$

Theorem 5.1 (Meromorphic Borel summation). *The series (5.2) is locally uniformly convergent away from its poles and is the Borel transform of*

$$\sum_{k \geq 1} \frac{B_{2k}}{(2k)!} \text{Li}_{2-2k}(e^{-s}) h^{2k-1}. \quad (5.3)$$

Its poles are simple and lie at

$$\xi = 4\pi^2 \ell j \pm 2\pi i \ell s, \quad \ell \geq 1, \quad j \in \mathbb{Z}. \quad (5.4)$$

The residue at an upper-half-plane pole is $-i/(2\pi\ell)$ and at a lower-half-plane pole is $+i/(2\pi\ell)$. For every $h, s > 0$,

$$\mathcal{A}_h(s) = \frac{\text{Li}_2(e^{-s})}{h} + \frac{1}{2} \log(1 - e^{-s}) + (\mathcal{L}\beta_s)(h). \quad (5.5)$$

Proof. In a neighborhood of $\xi = 0$, expand the two denominators in exponentials:

$$\beta_s(\xi) = 2 \sum_{\ell \geq 1} b_\ell^{-2} \sum_{m \geq 1} e^{-ms} \cos(m\xi/b_\ell). \quad (5.6)$$

The Taylor coefficient of ξ^{2k-2} is

$$\frac{2(-1)^{k-1}}{(2k-2)!} \sum_{\ell \geq 1} b_\ell^{-2k} \sum_{m \geq 1} m^{2k-2} e^{-ms},$$

which is exactly the Borel coefficient in (5.3) by (3.4). For a compact set of ξ values, the terms with large ℓ are uniformly $O(\ell^{-2})$. Only finitely many remaining terms can have poles in that compact set. This proves meromorphic continuation and local uniform convergence away from the poles. Solving the denominator equations and differentiating them gives (5.4) and the stated residues.

On the positive real axis the absolute value of (5.6) is at most $2\zeta(2)(2\pi)^{-2}(e^s - 1)^{-1}$. Hence its Laplace integral is absolutely convergent and summation may be interchanged with integration. Since

$$\int_0^\infty e^{-\xi/h} \cos(m\xi/b_\ell) d\xi = \frac{h}{1 + (hm/b_\ell)^2},$$

the result is the partial-fraction correction in (3.1), inserted into (2.4). This proves (5.5). $\square$

5.2 The multinomial Borel plane

At fixed $\tau > 0$ put

$$\beta_{\mathbf{a},\tau}(\xi) = \beta_{A\tau}(\xi) - \sum_j \beta_{a_j\tau}(\xi). \quad (5.7)$$

The correct Borel transform for the normalized $1/x$ series is

$$\mathcal{B}_{\mathbf{a},\tau}(\eta) = \tau \beta_{\mathbf{a},\tau}(\tau\eta). \quad (5.8)$$

The prefactor τ is essential: it accounts for replacing h by τ/x . The exact summation identity becomes

$$F_h(x) - \Phi_h(x) = \int_0^\infty e^{-x\eta} \mathcal{B}_{\mathbf{a},hx}(\eta) d\eta. \quad (5.9)$$

Theorem 5.2 (Universal nearest Borel singularities). *For every $\tau > 0$, the Taylor series of $\mathcal{B}_{\mathbf{a},\tau}$ at zero has radius of convergence exactly $2\pi a_*$. Its nearest singularities are*

$$\eta = \pm 2\pi i a_*, \quad (5.10)$$

with residues $+im_/(2\pi)$ at the upper pole and $-im_*/(2\pi)$ at the lower pole. They cannot cancel. Other candidate poles have the form*

$$\eta = \frac{4\pi^2 \ell j}{\tau} \pm 2\pi i \ell \lambda, \quad \lambda \in \{A, a_1, \dots, a_r\}. \quad (5.11)$$

At coincident poles the residues are added, and cancellation is possible. As $\tau \downarrow 0$, the meromorphic functions converge locally uniformly, away from the limiting poles, to

$$\mathcal{B}_{\mathbf{a},0}(\eta) = \sum_{\ell \geq 1} \left(\frac{2A}{b_\ell^2 A^2 + \eta^2} - \sum_j \frac{2a_j}{b_\ell^2 a_j^2 + \eta^2} \right). \quad (5.12)$$

Proof. Equation (5.11) follows by scaling (5.4). Every such point has modulus at least $2\pi\ell\lambda \geq 2\pi a_*$. Equality is possible only when $j = 0$, $\ell = 1$ and $\lambda = a_*$. The numerator parameter A is strictly greater than a_* . Thus exactly the m_* negative denominator contributions occur there, and their residues add rather than cancel. This proves the exact radius and the residues.

For the endpoint limit, each finite- ℓ term uses

$$\frac{\tau}{\exp\{\tau(\lambda \pm i\eta/b_\ell)\} - 1} \longrightarrow \frac{1}{\lambda \pm i\eta/b_\ell}.$$

On a fixed compact set and for sufficiently large ℓ , the real part of $\lambda \pm i\eta/b_\ell$ is at least $\lambda/2$. The corresponding terms are uniformly bounded by a constant times ℓ^{-2} , for all sufficiently small positive τ . Dominated convergence of the tail, together with uniform convergence of finitely many analytic terms away from their limiting poles, proves (5.12). $\square$

This gives a precise geometric interpretation of the endpoint normalization. The poles with $j \neq 0$ move off to infinity as $\tau \rightarrow 0$, while the nearest imaginary poles remain fixed. The limiting kernel is exactly the Borel kernel of the gamma-multinomial Stirling correction. Endpoint uniformity therefore corresponds to a controlled meromorphic limit, not to removal of the leading factorial divergence.

5.3 Large-order coefficients and what is not being claimed

For each fixed $\tau > 0$, the residue pair in Theorem 5.2 gives

$$D_k(\tau) \sim \frac{2m_*(-1)^k(2k-2)!}{(2\pi)^{2k}a_*^{2k-1}}, \quad k \rightarrow \infty. \quad (5.13)$$

The same equivalent holds at $\tau = 0$. To justify it directly, subtract the two nearest principal parts from (5.8). For fixed τ , there is a strictly larger disk free of the remaining singularities; Cauchy estimates make their coefficient contribution exponentially smaller in k . Expanding the two principal parts gives (5.13).

The relative error implicit in (5.13) is not asserted to be uniform as $\tau \rightarrow \infty$, when other poles approach the same modulus. The uniform truncation bound in Theorem 2.1 does not depend on such a coefficient equivalent and remains valid there.

The pole lattice determines residue contributions under an admissible contour deformation: a positively oriented small contour around a pole ω contributes $2\pi i \operatorname{Res}(\beta_{a,\tau}, \omega)e^{-\omega/h}$. This local statement is exact. A full global Stokes theorem would additionally specify lateral contours, growth estimates and all collision cases. We do not replace those requirements by a formal list of exponentials.

6 Uniform exponential accuracy and a sharp endpoint equivalent

Theorem 6.1 (An explicit least-term envelope). *Let $z = a_*x \geq 1$. Choose an odd integer N nearest to $2\pi z$, breaking an exact tie either way, and set $K = (N-1)/2$. Then for every $h \geq 0$,*

$$|R_K(h, x)| \leq E_K(x) < 3r\sqrt{z}e^{-2\pi z}. \quad (6.1)$$

In particular, for the central Gaussian ray,

$$|R_K(h, x)| < 6\sqrt{x}e^{-2\pi x}, \quad h \geq 0, \quad x \geq 1. \quad (6.2)$$

Proof. Because $a_j \geq a_*$, (2.19) and (3.4) imply

$$E_K(x) \leq \frac{2r\zeta(N+1)}{2\pi} \frac{N!}{(2\pi z)^N}. \quad (6.3)$$

Write $u = 2\pi z$ and $N = u + \delta$, $|\delta| \leq 1$. The elementary real Stirling upper bound $N! \leq \sqrt{2\pi N} (N/e)^N e^{1/(12N)}$ follows, for example, from Binet's positive first remainder [8]. The function $v \log(v/u) - v$ has value $-u$ and derivative zero at $v = u$, with second derivative $1/v$. Hence

$$N \log(N/u) - N \leq -u + \frac{1}{2(u-1)}.$$

It follows that

$$\frac{N!}{u^N} \leq \sqrt{2\pi(u+1)} \exp\left(-u + \frac{7}{12(u-1)}\right). \quad (6.4)$$

Here $u \geq 2\pi$, $N \geq 5$, and $\zeta(N+1) \leq \zeta(6) < 1.02$. Dividing the combination of (6.3) and (6.4) by $r\sqrt{z}e^{-2\pi z}$ bounds the ratio by

$$2(1.02)\sqrt{1 + \frac{1}{2\pi}} \exp\left(\frac{7}{12(2\pi-1)}\right) < 3.$$

This proves the stated constant without an unspecified asymptotic threshold. $\square$

6.1 Removing the polynomial loss uniformly

The preceding bound follows directly from the simple all-order expression E_K . A sharper lattice-moment estimate removes its factor $\sqrt{x}$.

Lemma 6.2 (A peak-sensitive lattice moment). *For every integer $N \geq 5$ and every $s > 0$,*

$$s^N \text{Li}_{1-N}(e^{-s}) \leq 3N^N e^{-N}. \quad (6.5)$$

Proof. If f is a nonnegative integrable unimodal function on $[0, \infty)$,

$$s \sum_{m \geq 1} f(ms) \leq \int_0^\infty f(t) dt + s \sup f. \quad (6.6)$$

Indeed, every positive superlevel set is an interval. The number of positive lattice points of spacing s in an interval is at most its length divided by s , plus one. Integrating this inequality over the superlevel height proves (6.6) by the layer-cake formula.

Apply it to $f(t) = t^{N-1}e^{-t}$. If $s \leq N+1$, the left side of (6.5) is at most

$$\Gamma(N) + (N+1)(N-1)^{N-1}e^{-(N-1)}.$$

After division by $N^N e^{-N}$, Stirling's upper bound makes the first term at most $\sqrt{2\pi/N} e^{1/(12N)} < 1.15$. The second term is

$$e(1+1/N)(1-1/N)^{N-1} \leq (1+1/N)e^{1/N} \leq 1.2e^{1/5} < 1.47.$$

Their sum is less than 3. If $s \geq N+1$, successive terms of $\sum m^{N-1}e^{-ms}$ have ratio at most $2^{N-1}e^{-s} \leq 16e^{-6} < 1/25$. Thus

$$s^N \text{Li}_{1-N}(e^{-s}) \leq \frac{25}{24} s^N e^{-s} \leq \frac{25}{24} N^N e^{-N} < 3N^N e^{-N}.$$

The last inequality uses the maximum of $t^N e^{-t}$ at $t = N$. $\square$

Theorem 6.3 (Uniform exponential bound without a polynomial loss). *With z , N and K chosen as in Theorem 6.1,*

$$\boxed{|R_K(h, x)| < 2re^{-2\pi z}, \quad h \geq 0, \quad z = a_*x \geq 1.} \quad (6.7)$$

Proof. Use the first-omitted-term inequality in (2.18) and discard the negative numerator contribution in its positive weight. For $h > 0$, Lemma 6.2 and (3.4) give

$$\begin{aligned} |R_K(h, x)| &< \frac{2\zeta(N+1)}{(2\pi)^{N+1}} h^N \sum_j \text{Li}_{1-N}(e^{-a_j h x}) \\ &\leq \frac{3r\zeta(N+1)}{\pi} \left(\frac{N}{2\pi z}\right)^N e^{-N}. \end{aligned}$$

The same estimate at $h = 0$ follows by continuity. Writing $u = 2\pi z$ and using $|N - u| \leq 1$ as before, the right side is at most

$$\frac{3r\zeta(6)}{\pi} \exp\left(-u + \frac{1}{2(u-1)}\right) < 2re^{-u}, \quad u \geq 2\pi.$$

The numerical constant 2 is deliberately simple rather than best possible. $\square$

The exponential here is a bound on optimally truncated remainders. The nearest Borel singularities are imaginary, not positive real. It would be incorrect to describe $e^{-2\pi a_* x}$ as a positive-real Stokes sector solely from this estimate. The following theorem gives the precise sharpness assertion.

Theorem 6.4 (Sharp endpoint remainder). *Fix a positive ray $\mathbf{a}$. Use the same nearest-odd rule for N and K as in Theorem 6.1. As $x \rightarrow \infty$,*

$$|R_K(0, x)| \sim \frac{m_*}{2\pi\sqrt{a_*x}} e^{-2\pi a_* x}. \quad (6.8)$$

Consequently, for this least-term truncation rule, no uniform estimate over $h \geq 0$ can replace the exponential rate $2\pi a_$ by $2\pi a_* + \varepsilon$, even allowing an arbitrary fixed polynomial prefactor in x .*

Proof. Let $b = 2\pi$ and $z = a_*x$. In (3.18), the terms with $\ell \geq 2$ are exponentially smaller in N : their denominator is larger, and the coefficient ratio to $\ell = 1$ is at most $\ell^{-(N-1)}$. The sum of these ratios is $O(2^{-N})$.

Within the $\ell = 1$ integral, the terms with $a_j > a_*$, and the numerator term with $A > a_*$, are likewise negligible. To see this quantitatively, write

$$I_N(y) = \int_0^\infty \frac{t^{N-1} e^{-yt}}{b^2 + t^2} dt = \frac{\Gamma(N)}{y^N} \mathbb{E} \frac{1}{b^2 + T_y^2}, \quad (6.9)$$

where T_y has gamma shape N and rate y . If $y = z$, its mean is $N/z \rightarrow b$ and its variance is $N/z^2 \rightarrow 0$. The bounded continuous integrand therefore has expectation tending to $1/(2b^2)$. In particular this expectation is bounded below by a positive constant for large x . For $y = a_j x > z$, its expectation is bounded above by $1/b^2$, so $I_N(a_j x)/I_N(z) = O((a_*/a_j)^N)$. The same applies to A .

Thus (3.18) is asymptotic in absolute value to

$$2m_* b^{-(N-1)} I_N(z) \sim \frac{m_* \Gamma(N)}{b^{N+1} z^N}. \quad (6.10)$$

Since $N = bz + O(1)$, Stirling's formula gives

$$\frac{\Gamma(N)}{(bz)^N} \sim \sqrt{\frac{2\pi}{bz}} e^{-bz}.$$

Substitute $b = 2\pi$ in (6.10) to obtain (6.8). The final assertion follows by evaluating any proposed stronger uniform bound at $h = 0$: an extra factor $e^{-\varepsilon x}$ cannot dominate the nonzero right-hand side of (6.8) up to a fixed polynomial. $\square$

6.2 Discrete resonances obstruct an endpoint-sized uniform prefactor

The $x^{-1/2}$ factor at $h = 0$ comes from a continuous saddle integral. At fixed $h > 0$, the exact remainder is a lattice sum. A lattice point can coincide with that saddle and remove the shrinking Gaussian-width factor. This gives a second sharpness result.

Theorem 6.5 (Resonant fixed- h remainder). *Fix an integer $m_0 \geq 1$ and set $h = 2\pi/m_0$. For a fixed positive ray, and the same nearest-odd rule as above,*

$$|R_K(2\pi/m_0, x)| \sim \frac{m_*}{2\pi m_0} e^{-2\pi a_* x}, \quad x \rightarrow \infty. \quad (6.11)$$

In particular, no estimate of the form $C_a g(x) e^{-2\pi a_ x}$ with $g(x) \rightarrow 0$ can hold uniformly for all $h \geq 0$ for this truncation rule. Combined with Theorem 6.3, this proves that the uniform remainder has precisely the order $e^{-2\pi a_* x}$ up to positive ray-dependent constant factors.*

Proof. Write $b = 2\pi$, $z = a_* x$ and $N = bz + \delta$, $|\delta| \leq 1$. The absolute version of (3.8) is

$$|R_K(h, x)| = 2h \sum_{m \geq 1} (mh)^{N-1} W_a(mhx) \sum_{\ell \geq 1} \frac{(b\ell)^{-(N-1)}}{(b\ell)^2 + (mh)^2}. \quad (6.12)$$

The $\ell \geq 2$ contributions are at most $\sum_{\ell \geq 2} \ell^{-(N-1)} = O(2^{-N})$ times the $\ell = 1$ contribution, since every weight is positive.

First retain only the $m_* e^{-a_* mhx}$ part of W_a . At $m = m_0$ the product mh equals b exactly, and its contribution for $\ell = 1$ is

$$\frac{m_* h}{b^2} e^{-bz} = \frac{m_*}{bm_0} e^{-bz}.$$

For any other m , put $v = m/m_0$. The ratio of its contribution to the m_0 contribution is

$$\frac{2v^{\delta-1}}{1+v^2} \exp\{-bz(v-1-\log v)\}. \quad (6.13)$$

The function $v - 1 - \log v$ is positive away from $v = 1$. On this fixed lattice its infimum for $m \neq m_0$ is positive; it grows at least linearly for large m . The prefactor is bounded, and summing the exponentially decaying tail shows that the sum of (6.13) is $O(e^{-cz})$ for some $c > 0$ depending on m_0 .

It remains to bound the terms in W_a with parameter $\lambda > a_*$, including its negative numerator term. Replacing the rational denominator in (6.12) by b^{-2} and using Lemma 6.2 bounds each such absolute contribution by

$$\frac{6}{b} \left(\frac{N}{b\lambda x} \right)^N e^{-N} = O\left(e^{-bz} (a_*/\lambda)^N\right).$$

This is negligible relative to the displayed m_0 contribution. It proves (6.11). The obstruction follows already from $m_0 = 1$. The lower bound on the supremum over h is supplied by that same value; Theorem 6.3 supplies the upper bound. $\square$

Remark 6.6. Both sharpness claims concern the specified Bernoulli truncation rule, not all possible numerical algorithms or hyperasymptotic corrections. Subtracting a resolved leading remainder can improve the error. The resonances occur at fixed positive h , so they do not contradict the endpoint equivalent or a possible improved estimate restricted to a sufficiently fine lattice $h = h(x) \rightarrow 0$. Such a restriction requires quantitative control of the lattice spacing relative to the saddle width.

7 Uniform conditioning and inverse certificates

An asymptotic formula for a forward function is not automatically an equally accurate inverse formula. A small derivative can amplify a residual. Here the multinomial structure gives a derivative bound valid uniformly in h .

Theorem 7.1 (Uniform lower bound on the inverse slope). *Set*

$$x_0 = a_*^{-1}, \quad c_a = A\psi(Ax_0 + 1) - \sum_j a_j \psi(a_j x_0 + 1) > 0, \quad (7.1)$$

where ψ is the digamma function. Then

$$F'_h(x) \geq F'_0(x) \geq c_a, \quad h \geq 0, \quad x \geq x_0. \quad (7.2)$$

For $a = (1, 1)$ one has $x_0 = 1$ and $c_a = 1$.

Proof. Differentiate (2.5) at fixed h and put

$$g_x(t) = \frac{\sum_j a_j e^{-a_j x t} - A e^{-A x t}}{e^t - 1}. \quad (7.3)$$

Then

$$F'_h(x) = 2\kappa h x + h \sum_{m \geq 1} g_x(hm), \quad g_x(0) = 2\kappa x. \quad (7.4)$$

For $x \geq x_0$, g_x is decreasing. Indeed it is the sum of the positive functions

$$a_j e^{-(a_j x + 1)t} \frac{1 - e^{-(A - a_j)x t}}{1 - e^{-t}}. \quad (7.5)$$

Here $(A - a_j)x \geq 1$. For any $v \geq 1$,

$$\frac{d}{dt} \log \frac{1 - e^{-vt}}{1 - e^{-t}} = \frac{v}{e^{vt} - 1} - \frac{1}{e^t - 1} \leq 0,$$

because $v \mapsto v/(e^{vt} - 1)$ is decreasing. Thus every summand in (7.5) is decreasing.

The lower Riemann sum inequality is

$$h \sum_{m \geq 1} g_x(hm) \geq \int_0^\infty g_x(t) dt - h g_x(0).$$

The subtracted term cancels the quadratic derivative in (7.4). The remaining integral is $F'_0(x)$, by the elementary integral representation of digamma differences. Hence $F'_h(x) \geq F'_0(x)$.

It remains to show that F'_0 is strictly increasing and positive. The trigamma integral gives

$$F''_0(x) = \int_0^\infty s e^{-xs} \left(\frac{1}{e^{s/A} - 1} - \sum_j \frac{1}{e^{s/a_j} - 1} \right) ds. \quad (7.6)$$

For fixed $s > 0$, the function $u \mapsto [u(e^{s/u} - 1)]^{-1}$ is strictly increasing. Since $a_j < A$,

$$\sum_j \frac{1}{e^{s/a_j} - 1} < \sum_j \frac{a_j}{A} \frac{1}{e^{s/A} - 1} = \frac{1}{e^{s/A} - 1}.$$

Therefore $F_0''(x) > 0$. Also $F_0'(0) = 0$, since $A = \sum_j a_j$. This proves (7.1) and (7.2). In the central case, $2\psi(3) - 2\psi(2) = 1$. $\square$

For $h > 0$, the numerator in (7.3) is positive for every $x > 0$. Thus F_h is increasing already below x_0 ; the threshold is used only for the uniform numerical slope bound. Moreover $F_h(0) = 0$ by continuous extension of (2.5), and $F_h(x) \rightarrow \infty$. The same statements hold for F_0 . Each has an unambiguous inverse on positive logarithmic targets.

Theorem 7.2 (Residual-to-inverse transport). *Let $L \geq F_h(x_0)$, and let x_* be the unique solution of $F_h(x_*) = L$. For any $s \geq x_0$, put*

$$\rho = T_K(h, s) - L, \quad e = E_K(s). \quad (7.7)$$

Then

$$|x_* - s| \leq \frac{|\rho| + e}{c_a}. \quad (7.8)$$

When $\rho = 0$, the sharper one-sided enclosures are

$$s \leq x_* \leq s + e/c_a, \quad K \text{ even}, \quad (7.9)$$

$$\max\{x_0, s - e/c_a\} \leq x_* \leq s, \quad K \text{ odd}. \quad (7.10)$$

One may replace e throughout by the smaller next-term bound in (2.18). If K is chosen by the least-term rule at s , then

$$|x_* - s| \leq \frac{|\rho| + 2re^{-2\pi a_* s}}{c_a}. \quad (7.11)$$

Proof. The exact residual is $F_h(s) - L = \rho + R_K(h, s)$. Theorem 2.1 bounds its magnitude by $|\rho| + e$. Both s and x_* lie in $[x_0, \infty)$, so the mean value theorem and Theorem 7.1 give (7.8). When $\rho = 0$, the sign in (2.18) determines which side of s contains x_* . Finally use Theorem 6.3. $\square$

7.1 Existence and monotonicity of truncated inverse equations

For an odd K , (2.20) says $\partial_x R_K < 0$. Consequently

$$\partial_x T_K(h, x) = F_h'(x) - \partial_x R_K(h, x) \geq c_a \quad (x \geq x_0). \quad (7.12)$$

Also $T_K(h, x_0) < F_h(x_0)$ and $T_K(h, x) \rightarrow \infty$. Thus an odd truncation has a unique root $T_K(h, s) = L$ in $[x_0, \infty)$ for every target in Theorem 7.2. This is a globally well-conditioned approximate inverse on that range.

For an even K , monotonicity is guaranteed wherever $(2K + 2)E_K(x)/x < c_a$. This is an explicit, checkable threshold. The general residual certificate does not require solving any truncated equation exactly, and therefore remains useful for Newton iterates, interval bisection, or a formal inverse series generated by another method.

7.2 Using formal reversion without confusing it with certification

Suppose a chosen core Φ is locally invertible, with $\Phi(s) = L$, and write $\Delta = F - \Phi$. Classical inverse perturbation gives, formally in an auxiliary parameter multiplying Δ ,

$$x = s + \sum_{j \geq 1} \frac{(-1)^j}{j!} \left(\frac{1}{\Phi'(s)} \frac{d}{ds} \right)^{j-1} \frac{\Delta(s)^j}{\Phi'(s)}. \quad (7.13)$$

The repository already develops this kind of reversion calculus [2, 1]. The first terms are

$$x = s - \frac{\Delta}{\Phi'} + \frac{\Delta \Delta'}{(\Phi')^2} - \frac{\Delta^2 \Phi''}{2(\Phi')^3} + \cdots. \quad (7.14)$$

In the present setting a finite candidate produced from this identity can be substituted into T_K and certified by (7.8). We do not assert convergence of a Bernoulli-based infinite inverse series. The separation between coefficient production and error certification is precisely what makes the uniform result usable at the endpoint.

8 Modular sectors, finite evaluation, and integer decisions

8.1 Two different exponential phenomena

The modular correction has the convergent expansion

$$M(h) = \sum_{n \geq 1} \sigma_{-1}(n) e^{-4\pi^2 n/h}, \quad \sigma_{-1}(n) = \sum_{d|n} \frac{1}{d}. \quad (8.1)$$

It follows by expanding $-\log(1 - \hat{Q}^j)$ and grouping terms with the same product of indices, where $\hat{Q} = e^{-4\pi^2/h}$. Theorem 5.1 sums the finite-size Bernoulli tail exactly. The full forward function is that Borel sum plus the separately specified modular contribution $dM(h)$ and the elementary phase and amplitude.

In the crossover $x = \tau/h$, the uniform least-term envelope has exponential size $e^{-2\pi a_* \tau/h}$, whereas the first modular sector has size $e^{-4\pi^2/h}$. Their exponents meet at $\tau = 2\pi/a_*$. This comparison is useful for budgeting guaranteed errors; it is not an assertion that an actual remainder is asymptotic to its upper envelope for all τ . In particular, keeping the modular exponential while leaving an uncontrolled algebraic tail does not establish beyond-all-orders accuracy.

8.2 A finite evaluation budget for the modular factor

For $0 < q < 1$,

$$-\log \prod_{j \geq 1} (1 - q^j) = \sum_{m \geq 1} \frac{q^m}{m(1 - q^m)}. \quad (8.2)$$

If this sum is cut after $m = J$, its positive remainder is bounded by

$$\frac{q^{J+1}}{(J+1)(1-q)^2}. \quad (8.3)$$

Indeed $m \geq J+1$ and $1 - q^m \geq 1 - q$, after which a geometric sum finishes the estimate. For small h , use $q = \hat{Q}$. For large h , use

$$M(h) = -\log \mathcal{P}(h) + \frac{1}{2} \log \frac{2\pi}{h} - \frac{\pi^2}{6h} + \frac{h}{24}, \quad (8.4)$$

so that a product with parameter at least 2π is always available. Thus retaining M exactly in the theorem does not require an infinite computation in a finite certificate.

8.3 Coefficient evaluation and cancellation control

For a nonnegative integer p ,

$$\mathrm{Li}_{-p}(z) = \frac{zP_p(z)}{(1-z)^{p+1}}, \quad (8.5)$$

where for $p \geq 1$ the coefficients of P_p are Eulerian numbers; $p = 0$ is handled as $z/(1-z)$. The recurrence

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = (k+1) \left\langle \begin{matrix} n-1 \\ k \end{matrix} \right\rangle + (n-k) \left\langle \begin{matrix} n-1 \\ k-1 \end{matrix} \right\rangle \quad (8.6)$$

gives exact integer coefficient arrays. Thus every correction D_k is computable by rational operations and real exponentials once the Bernoulli number is known. Near zero one should evaluate $1 - e^{-s}$ with `expm1`, or use the removable-singularity expansion directly.

The remaining dilogarithms can be evaluated by their defining series; for $0 \leq z < 1$ the tail after J terms is bounded by

$$\sum_{m>J} \frac{z^m}{m^2} \leq \frac{z^{J+1}}{(J+1)^2(1-z)}. \quad (8.7)$$

Near $z = 1$ this is inefficient. It is preferable to evaluate the balanced phase via (3.16), its local Taylor series, or standard dilogarithm functional identities. Fixed working precision without cancellation control is not a certificate, even when the analytic truncation error is tiny.

8.4 From a real inverse interval to an integer answer

When the ray parameters are positive integers, the lattice values at integer x form an increasing sequence. Let $x_* = F_h^{-1}(\log Y)$. The least index whose value is at least Y is $\lceil x_* \rceil$. A certified interval $[u, v]$ determines that index whenever it lies in a single ceiling cell $(n-1, n]$. If an endpoint touches an integer, the convention at equality must be respected; exact or validated comparison with that lattice value resolves the boundary.

This is an application of the repository's general distinction between real inversion and staircase inversion [2], not a replacement for it. An interval of width less than one alone does not suffice: it may still straddle an integer threshold.

9 Verification and reproducibility

The accompanying `verify.py` implements the normalized coefficients, core, analytic bounds, least-term rule and inverse enclosure formulas using `mpmath`. The reference forward values on integral rays are computed from the finite Gaussian product, independently of the asymptotic expansion; at $h = 0$ the reference is the gamma quotient. This avoids validating an expansion only against a second evaluation of the same truncated expression.

The recorded run used 240 decimal digits. It checked 812 instances of the signed forward and first-omitted-term inequalities, including $h = 0$, $h = 10^{-8}$, crossover values and $h = 10$, for rays $(1, 1)$, $(1, 2)$, $(1, 1, 1)$ and $(1, 3, 5)$ and indices from 1 to 20. It also checked 24 inverse enclosures, 36 resonant least-term remainders, 48 peak-sensitive lattice moments, and auxiliary moment inequalities. Every assertion passed. The largest observed ratio $|R_K|/E_K$ was approximately 0.92747955.

At $h = 0$, $\mathbf{a} = (1, 1)$ and $x = 40$, the ratio of the measured absolute error to the sharp equivalent in (6.8) is approximately 1.00448044. The nearest-odd rule changes discretely with x , so the finite- x ratio need not approach one monotonically. For the fixed resonance $h = 2\pi/3$

h	x	K	$ F - T_K $	$E_K(x)$
0.1	2	6	7.8486×10^{-7}	2.0345×10^{-5}
0	5	15	3.365×10^{-15}	2.0289×10^{-13}
0.00000001	5	15	3.365×10^{-15}	2.0289×10^{-13}
0.1	5	15	3.365×10^{-15}	2.0289×10^{-13}
2	5	15	3.3487×10^{-15}	2.0289×10^{-13}
10	5	15	6.2724×10^{-17}	2.0289×10^{-13}
0.1	10	31	5.2196×10^{-29}	6.5431×10^{-27}
0.1	20	62	1.9207×10^{-56}	4.758×10^{-54}

Table 1: Central-ray least-term checks. The target is the exact finite-product value at the displayed integral x (the gamma value for $h = 0$). Entries are rounded numerical observations, not outward-rounded interval endpoints.

at $x = 40$, the ratio to (6.11) is approximately 1.00000514; for $h = 2\pi$ it agrees with one to the 14 displayed digits.

For a simple inverse example, use the central ray, $h = 0$, and target $L = F_0(5) = \log 252$. Solving $T_3(0, s) = L$ gives $s - 5 \approx 1.1141787 \times 10^{-8}$; the one-sided analytic enclosure has width about $1.06666665 \times 10^{-7}$. For the same index and $h = 10^{-4}$, the observed displacement is 1.1137474×10^{-8} . This illustrates the continuous endpoint transition without changing the certification method.

What these checks establish. They are high-precision consistency tests for formulas and signs. They are not proofs of the theorems, not interval arithmetic, and not Lean kernel checks. The package’s numerical enclosure routine prints high-precision approximations to analytic enclosure endpoints; those printed endpoints are not themselves machine-certified until elementary evaluations and rounding are validated. The mathematical certificate is Theorem 7.2.

Reproduction. From the package directory run

```
python -m pip install -r requirements.txt
python verify.py
pdflatex -interaction=nonstopmode -halt-on-error uniform_q_multinomial_transseries.
tex
pdflatex -interaction=nonstopmode -halt-on-error uniform_q_multinomial_transseries.
tex
```

The Python run regenerates `verification_results.json` and `numerical_table.tex`. The build does not depend on repository-local notation files or external images. No repository files were modified.

10 A proposed formalization route

The proof divides naturally into a finite algebraic layer, a real-analysis layer, and an optional complex-analysis layer. This makes the uniform theorem a practical next target for the repository’s existing residual-certificate architecture, without requiring a complete formal datatype for transseries.

At the finite layer, formalize the geometric identity that produces (3.3); the Bernoulli–zeta coefficient identity may be supplied through a separate analytic theorem. Establish positivity

of W_a and $-W'_a$ using only the strict inequalities $a_j < A$. The moment estimate in Lemma 3.1 is an independent reusable lemma.

At the real-analysis layer, formalize the convergent Bose partial fractions, the dominated summations in (3.8), and the exact modular identity as separately named dependencies. Prove Theorem 2.1 first for $h > 0$, then add the Binet endpoint. A finite truncation object with fields `value`, `errorBound` and `errorSign` would already support all inverse certificates. It should not be represented as a convergent infinite Bernoulli series.

The slope theorem is particularly useful independently of transseries. Its proof needs decreasing-function Riemann sums and a positive integral representation of the trigamma combination. Once it is available, the inverse theorem is a direct mean-value argument; the repository already has related residual-transport interfaces [2].

The Borel pole lattice and the sharp endpoint equivalent are additional analytic layers. They are not prerequisites for the finite-order inverse certificate. Separating them avoids inflating a useful elementary theorem's formal dependency graph. No Lean implementation or successful Lean build is claimed in this package.

11 Further research questions and topics

The following questions are proposed next steps, not claims that each is an established open problem in the published literature. Their exact status would require a focused literature review before being advertised that way.

Q1. The lattice-to-integral transition of the least remainder. Theorem 6.5 disproves an $x^{-1/2}$ prefactor uniform in all h . Determine a full transition formula when $h\sqrt{a_*x}$ is bounded, including the fractional position of the saddle $2\pi/h$ relative to the integer lattice. A Gaussian lattice sum suggests theta-function factors, but a uniform theorem must control both the tails and the next partial-fraction poles. This would interpolate between (6.8) and the discrete resonance law rather than incorrectly extrapolating either one.

Q2. A uniform first hyperasymptotic correction. Resolve the $\ell = 1$, smallest-parameter contribution of (3.8) explicitly, including its dependence on h , and then bound the remainder after subtracting it. The next relevant scales may come from $2a_*$, the second-smallest ray parameter, and off-axis Borel poles. A proof should distinguish those mechanisms instead of taking the minimum of a guessed list of exponents.

Editorial note (ProveIt, 2026-09-29). Q1 and Q2 are taken up by the theta-resolved package [9], written from a saved copy of this article. For Q1, its theorems `thm:theta` and `thm:optimal` give an all-order theta-lattice expansion of the exact remainder and its minimum over all orders, uniformly for $h \geq 0$ with $h\sqrt{a_*x}$ bounded, including the lattice phase $\{2\pi/h\}$; `thm:fixed` and `thm:overlap` give the fixed nonresonant-mesh law and match it to the theta law. For Q2, its `thm:hierarchy` resolves the first L partial-fraction pole components exactly (the whole weight, not only the smallest-ray term) and moves the critical-mesh action to $2\pi(L+1)a_*x$. Both answers are for a fixed ray and a fixed number of resolved poles; that package does not claim general resurgent completion, complex sectors, a growing pole count or a complete resonance boundary layer, and the questions are not marked resolved beyond that scope.

Q3. Growing or degenerating rays. The theorem fixes r and the positive a_j . Determine useful uniform statements when $r \rightarrow \infty$, when $a_* \rightarrow 0$, or when several parameters

coalesce at an x -dependent rate. The explicit sums in E_K expose the possible loss of constants, but do not by themselves provide an optimal joint scaling theorem.

- Q4. Positive signed product ratios beyond multinomials.** For a general finite signed exponential weight $W(s) = \sum_{\nu} c_{\nu} e^{-\lambda_{\nu} s}$, characterize tractable sufficient conditions for $W \geq 0$ and $-W' \geq 0$. Such conditions would transfer the present remainder and conditioning arguments to wider balanced q -gamma quotients. Necessity of these conditions for alternating remainder laws is a separate, more delicate question.
- Q5. Complex sectors and pole pinching.** Develop uniform lateral Laplace estimates when τ approaches zero through complex directions. The real theorem exploits positivity; it does not cover contours pinched by moving poles. The meromorphic limit in Theorem 5.2 supplies explicit local data for this analysis.
- Q6. Complete collision and Stokes-factor classification.** For commensurable ray parameters, classify all cancellations among the poles in (5.11), including residues at nonzero real translates. Then prove global Stokes formulas with specified contours and convergence regions. The nearest pole cannot cancel, but later singularities can.
- Q7. Uniform explicit inverse expansions.** Turn (7.13) into a single coefficient calculus with explicit derivative bounds uniform in $h \geq 0$. The present article certifies finite inverse candidates; it does not prove a complete resurgent expansion of the inverse across every endpoint and complex sector.
- Q8. Validated fast computation.** Implement the phase and coefficient evaluation with directed rounding, adaptive precision and cancellation-aware formulas. Compare direct products, q -gamma evaluation, normalized asymptotics and hyperasymptotics. The goal is a fully verified enclosure routine, not merely more digits from an unvalidated evaluator.
- Q9. Summed q -multinomials and theta transitions.** Extend the approach from one multinomial ray to the generalized Galois numbers and root-lattice theta sectors treated elsewhere in the companion. Summing over a moving lattice introduces a width transition absent from the fixed-ray problem. It should not be assumed that the same smallest-parameter law survives unchanged.
- Q10. A formal optimal-truncation certificate.** Formalize the complete chain from the signed kernel through the explicit Bernoulli bound and nearest-odd rule to the inverse enclosure. The sharp endpoint equivalent can be a second-stage target. This would convert the repository's supporting least-term results into an assembled theorem for a nontrivial analytic family.

12 Conclusion

The explicit repository gap at $hx = 0$ can be closed without a new abstract transseries field. The decisive step is an exact signed kernel identity that survives the multinomial combination. Normalizing in powers of $1/x$ removes the artificial endpoint singularities from the coefficient functions and produces an error bound uniform for every $h \geq 0$.

The Borel analysis explains the resulting exponential scale. The nearest singularities are fixed at $\pm 2\pi i a_*$ in the normalized Borel plane; the gamma endpoint retains them and gives the sharp least-term equivalent. The peak-sensitive moment bound removes the polynomial loss uniformly, while fixed lattice resonances prove that a vanishing uniform prefactor is impossible

for the specified truncation rule. The modular Euler-product sectors are separate data and must remain in the forward model when exponentially small accuracy is requested. Finally, the uniform derivative comparison turns the forward expansion into a practical inverse certificate, including the classical endpoint and the fixed- q regime.

These results are a concrete extension of the repository's transseries program. They do not settle general resurgence or global inversion questions, and their worldwide novelty is not certified. The strongest claims made here are the precisely stated theorems with the proofs above.

A A compact theorem and dependency ledger

Result	Content	Main inputs
Theorem 2.1	All- h signed expansion, explicit remainder and derivative bounds	Bose partial fractions, positive weight, moment lemma, eta identity, Binet endpoint
Theorem 5.1	Exact single-tail Borel transform and Laplace sum	Geometric series, cosine transform, normally convergent meromorphic sums
Theorem 5.2	Universal nearest poles and endpoint kernel	Explicit Borel formula, residue addition, dominated convergence
Theorem 6.1	Explicit uniform least-term envelope	Bernoulli-zeta identity and real Stirling bound
Theorem 6.3	Uniform $O(e^{-2\pi a_* x})$ error	Peak-sensitive lattice moment
Theorem 6.5	Sharp fixed- h resonances and the obstruction to a vanishing uniform prefactor	Discrete positive remainder, saddle separation
Theorem 6.4	Sharp endpoint equivalent	Exact Binet remainder, gamma concentration, Stirling formula
Theorem 7.1	$F'_h \geq F'_0 \geq c_a$	Decreasing Riemann sums, digamma and trigamma integrals
Theorem 7.2	One-sided inverse enclosures	Uniform forward error and exact slope bound

B Conventions and common failure modes

The expansion variable in Theorem 2.1 is $1/x$; hx is retained inside the coefficient functions. The Borel variable ξ initially refers to h , whereas $\eta = \xi/\tau$ refers to $1/x$. Confusing these variables changes the apparent nearest action.

The approximation T_K contains K Bernoulli terms, ending at $x^{-(2K-1)}$. The first omitted power is $N = 2K + 1$. Choosing $N \approx 2\pi a_* x$ is therefore consistent with $K \approx \pi a_* x$, not $K \approx 2\pi a_* x$.

The signs are also worth recording. R_K has sign $(-1)^{K+1}$, so odd truncations lie below the exact forward function and yield upper inverse approximations. Even truncations lie above it and yield lower inverse approximations whenever the relevant truncated root is used. The derivative of R_K has the opposite sign.

The interpolation is fixed by (2.5); a different real interpolation of the same sequence can have a different inverse between integer points. All staircase conclusions must use that convention or an explicit comparison theorem.

Finally, a list of high-precision decimal values is not an interval proof. A computationally certified inverse needs both the analytic truncation bound and validated evaluation of the finite expression and its residual.

References

- [1] Vladimir Reshetnikov, *ProveIt repository: Combinatorial Transseries and Their Inverses*, source `Combinatorial_Transseries_Inverses.tex`, inspected at commit 04e06e032dff1966513bfba316966d52db3646b4, 29 September 2026. Full repository path and inspected labels are recorded in `source_provenance.md`. Repository: <https://github.com/VladimirReshetnikov/ProveIt>. Editorial note (ProveIt, 2026-09-29): the path in `source_provenance.md` is the pre-split one; the companion is now `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`.
- [2] Vladimir Reshetnikov, *ProveIt repository: Series and transseries*, `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries/README.md`, inspected 29 September 2026, source blob 516aaa30e84cb0036a00f803cb6e8e3026febb8e. Editorial note (ProveIt, 2026-09-29): that path no longer exists; the group README is now `Analysis/Transseries/docs/series-and-transseries/README.md`.
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- [9] [Editorial addition, ProveIt, 2026-09-29.] The five packages of the $q \rightarrow 1$ merge unit in the ProveIt series-and-transseries collection, `Analysis/Transseries/docs/series-and-transseries/Uniform_q_Multinomial_Certified_Inversion/uniform_q_multinomial_transseries.tex`; `Uniform_Resurgent_Crossover_Gaussian_Binomials/article.tex`; `Certified_Inversion_q_to_1_Transition/q_transition.tex`; `Gamma_Core_q_to_1_Crossover/Gamma_Core_Transseries.tex`; `Theta_Resolved_Optimal_Truncation_q_Multinomial/article.tex`. Filed in batches 45 and 46 of `docs/incoming/README.md`; unrefereed.
- [10] [Editorial addition, ProveIt, 2026-09-29.] ProveIt, *Gaussian Coefficient Calculus*, `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Gaussian_Coefficient_Calculus/Gaussian_Coefficient_Calculus.tex`, Theorem q3:thm:double-scaling (line 3942) and its domain q3:eq:double-scaling-domain.
- [11] [Editorial addition, ProveIt, 2026-09-29.] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion* (canonical volume), `Analysis/Transseries/docs/series-and-transseries/Transseries_And_`

Inversion/transseries_and_inversion.tex, Theorem p0:thm:optimal-truncation (line 39360).

- [12] [Editorial addition, ProveIt, 2026-09-29.] ProveIt Lean library: `Fabius.exists_eq_in_residual_interval` in `Analysis/FabiusFunction/Lean/FabiusFunction/MeanValueBracket.lean` and `Fabius.transport_bound` (the crosswalk of the canonical volume's `p0:eq:transport-bound`) in `Analysis/FabiusFunction/Lean/FabiusFunction/RemainderTransport.lean`.